

Experiment No. - 07

AIM:- To demonstrate the functions of Cisco Discovery Protocol.

● **STEPS :**

Step 1: Setup Network Topology

1. **Open Cisco Packet Tracer** and create a new project.
2. **Add Devices:**
 - **Two Cisco Routers** (Router0 and Router1).
 - **One Straight-Through Cable** to connect them.
3. **Connect the routers:**
 - Use a **Serial DCE cable** if using serial interfaces.
 - Use a **Copper Cross-over cable** if using FastEthernet or GigabitEthernet.

Step 2: Configure IP Addresses on Routers (Optional, for CDP Detail Testing)

1. Click **Router0**, go to **CLI** and configure:

```
Router0> enable
Router0# configure terminal
Router0(config)# interface FastEthernet0/0
Router0(config-if)# ip address 192.168.1.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
```

2. Click **Router1**, go to **CLI** and configure:

```
Router1> enable
Router1# configure terminal
Router1(config)# interface FastEthernet0/0
Router1(config-if)# ip address 192.168.1.2 255.255.255.0
Router1(config-if)# no shutdown
Router1(config-if)# exit
```

Step 3: Verify CDP Between Routers

1. Check if CDP is Running

- By default, CDP is **enabled** on Cisco routers.
- To verify if CDP is enabled, enter:

```
Router0# show cdp
```

2. View CDP Neighbors

- Run the following command on **Router0**:

```
Router0# show cdp neighbors
```

Expected Output (example):

```
sql
CopyEdit
Device ID      Local Intf      Holdtime      Capability      Platform
Port ID
Router1        Fa0/0           178           R               1941
Fa0/0
```

- This shows Router1 is connected to Router0 via FastEthernet0/0.

- Run the same command on **Router1**:

```
Router1# show cdp neighbors
```

3. View Detailed CDP Information

- To get **IP addresses, software version, and platform details** of the neighbor, run:

```
Router0# show cdp neighbors detail
```

Example Output:

```
Device ID: Router1
IP address: 192.168.1.2
Platform: Cisco 1941
Software Version: 15.2(4)M6
```

4. Disable CDP (Optional)

- **On a Specific Interface:**

```
Router0(config)# interface FastEthernet0/0
Router0(config-if)# no cdp enable
```

- **Globally Disable CDP:**

```
Router0(config)# no cdp run
```

Step 4: Testing CDP Changes

1. Run `show cdp neighbors` before and after disabling CDP on an interface.
2. Observe that the router **disappears** from the neighbor list.
3. Enable CDP again if needed:

```
Router0(config)# cdp run
```

Conclusion

- CDP helps discover directly connected Cisco devices.
- Useful for troubleshooting without knowing IP details.
- Can be disabled for security reasons in untrusted networks.